Power System Protection II

EE 511

EIGENVALUES AND EIGENVECTORS

Consider an arbitrary square matrix $\bar{A}$ If a relation exists so that.

$$\bar{A}\bar{x} = \lambda\bar{x}$$

where λ is a scalar called an eigenvalue, characteristic value, or root of the matrix $\bar{A}$, and x is a vector called the eigenvector or characteristic vector of $\bar{A}$.
Then, there are n eigenvalues and corresponding n sets of eigenvectors associated with an arbitrary matrix $\bar{A}$ of dimensions n×n. The eigenvalues are not necessarily distinct, and multiple roots occur.

$$\left[\bar{A} - \lambda I\right][\bar{x}] = 0$$

where I the is identity matrix

$$\begin{bmatrix} a_{11}-\lambda & a_{12} & a_{13} & \dots & a_1n \\ a_{21} & a_{22}-\lambda & a_{23} & \dots & a_2n \\ \dots & \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn}-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \dots \\ 0 \end{bmatrix}$$

This represents a set of homogeneous linear equations. Determinant $|A - \lambda I|$ must be zero as $\bar{x} \neq 0$.

$$\left|\bar{A} - \lambda I\right| = 0$$

This can be expanded to yield an nth order algebraic equation:

$$a_n\lambda^n + a_n - I\lambda^n - 1 + \ldots + a_1\lambda + a_0 = 0, \text{ i.e.,}$$

$$(\lambda_1 - a_1)(\lambda_2 - a_2)\ldots(\lambda_n - a_n) = 0$$

Equations $\left|\bar{A} - \lambda I\right| = 0$ is called the characteristic equations of the matrix $\bar{A}$. The roots $\lambda_1, \lambda_2, \lambda_3, \ldots, \lambda_n$ are the eigenvalues of matrix $\bar{A}$. The eigenvector $\bar{x}_i$ corresponding to $\bar{\lambda}_j$ is found from $\left[\bar{A} - \lambda I\right][\bar{x}] = 0$

Find the characteristic equation of the matrix:

$$\bar{A} = \begin{bmatrix} 1 & 4 & 2 \\ 3 & 2 & -2 \\ 1 & -1 & 2 \end{bmatrix}$$

The characteristic equation is given by

$$\begin{vmatrix} 1-\lambda & 4 & 2 \\ 3 & 2-\lambda & -2 \\ 1 & -1 & 2-\lambda \end{vmatrix} = 0$$

Expanding, the characteristic equation is

$$\lambda^3 - 5\lambda^2 - 8\lambda + 40 = 0$$

Similarity Transformation

Consider a system of linear equations:

$$\bar{A}\bar{x} = \bar{y}$$

A transformation matrix C can be introduced to relate the original vectors $\bar{x}$ and $\bar{y}$ to new sets of vectors $\bar{x}_n$ and $\bar{y}_n$ so that

$$\bar{x} = \bar{C}\bar{x}_n$$

$$\bar{y} = \bar{C}\bar{y}_n$$

$$\bar{A}\bar{C}\bar{x}_n = \bar{C}\bar{y}_n$$

$$\bar{C}^{-1}\bar{A}\bar{C}\bar{x}_n = \bar{C}^{-1}\bar{C}\bar{y}_n$$

$$\bar{C}^{-1}\bar{A}\bar{C}\bar{x}_n = \bar{y}_n$$

This can be written as

$$\bar{A}_n\bar{x}_n = \bar{y}_n$$

$$\bar{A}_n = \bar{C}^{-1}\bar{A}\bar{C}$$

$\bar{A}_n\bar{x}_n = \bar{y}_n$ is distinct from $\bar{A}\bar{x} = \bar{y}$. The only restriction on choosing $\bar{C}$ is that it should be nonsingular

If $\bar{C}$ is defined as follows

$$\bar{C} = [x_1, x_2, \ldots, x_n]$$

where $\bar{x}_i$ are the eigenvectors of the matrix $\bar{A}$

$$\bar{C} = [x_1, x_2, \ldots, x_n]$$

$$\bar{C}^{-1}\bar{A}\bar{C} = \bar{C}^{-1}\bar{A}[x_1, x_2, \ldots, x_n]$$

$$\bar{C}^{-1}[\bar{A}x_1, \bar{A}x_2, \ldots, \bar{A}x_n]$$

$$= \bar{C}^{-1}[\lambda_1 x_1, \lambda_2 x_2, \ldots, \lambda_n x_n]$$

$$= C^{-1}[x_1, x_2, \ldots, x_n]\begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \cdot & \\ & & & \lambda_n \end{bmatrix}$$

$$= \bar{C}^{-1}\bar{C}\begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \cdot & \\ & & & \lambda_n \end{bmatrix}$$

$$= \bar{\lambda}$$

Thus, $\bar{C}^{-1}\bar{A}\bar{C}$ is reduced to a diagonal matrix $\bar{\lambda}$, called a spectral matrix. Its diagonal elements are the eigenvalues of the original matrix $\bar{A}$. The new system of equations is an uncoupled system. Equations

$$\bar{A}\bar{x} = \bar{y}$$
$$\bar{A}_n = \bar{C}^{-1}\bar{A}\bar{C}$$

constitute a similarity transformation of matrix $\bar{A}$.

The matrices $\bar{A}$ and $\bar{A}_n$ have the same eigenvalues and are called similar matrices. The transformation matrix $\bar{C}$ is nonsingular.

$$\text{Let } \bar{A} = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

Its characteristics equation is

$$\left|\bar{A} - \lambda I\right| = 0$$

$$\lambda^3 + \lambda^2 - 21\lambda - 45 = 0$$

$$(\lambda - 5)(\lambda - 3)(\lambda - 3) = 0$$

The eigenvector is found by substituting the eigenvalues:

$$\begin{bmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

As eigenvectors are not unique, by assuming that $x_3 = 1$, and solving, one eigenvector is

$$\begin{matrix} -1 \\ -2 \\ 1 \end{matrix}$$

Similarly, other eigenvectors can be found. A matrix formed of these vectors is

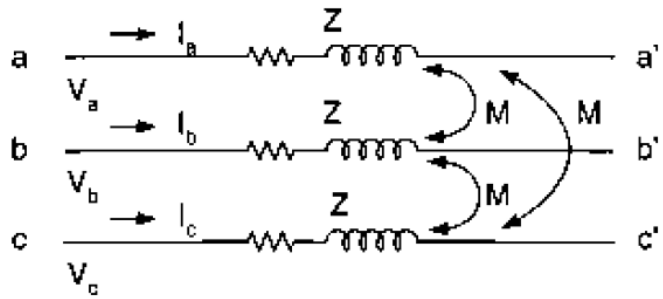
$$\bar{C} = \begin{bmatrix} -1 & 2 & 3 \\ -2 & -1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

and the diagonalization is obtained:

$$\bar{C}^{-1}\bar{A}\bar{C} = \begin{bmatrix} 5 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{bmatrix}$$

This contains the eigenvalues as the diagonal elements.

Decoupling a Three-Phase Symmetrical System



(a)

$$\begin{bmatrix} Z_{aa} & Z_{ab} & Z_{ac} \\ Z_{ba} & Z_{bb} & Z_{bc} \\ Z_{ca} & Z_{cb} & Z_{cc} \end{bmatrix}$$

Assume that the line is *perfectly symmetrical*. This means all the mutual impedances, i.e., $Z_{ab} = Z_{ba} = M$ and all the self-impedances, i.e., $Z_{aa} = Z_{bb} = Z_{cc} = Z$ are equal. This reduces the impedance matrix to

$$\begin{bmatrix} Z & M & M \\ M & Z & M \\ M & M & Z \end{bmatrix}$$

$$\begin{vmatrix} Z-\lambda & M & M \\ M & Z-\lambda & M \\ M & M & Z-\lambda \end{vmatrix} = 0$$

$$\begin{aligned} \lambda &= Z + 2M \\ &= Z - M \\ &= Z - M \end{aligned}$$

The eigenvectors can be found by making $\lambda = Z + 2M$ and then $Z - M$. Substituting $\lambda = Z + 2M$:

$$\begin{bmatrix} Z-(Z+2M) & M & M \\ M & Z-(Z+2M) & M \\ M & M & Z-(Z+2M) \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = 0$$

This can be reduced to

$$\begin{bmatrix} -2 & 1 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = 0$$

$$\begin{bmatrix} -2 & 1 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = 0$$

This give $X_1 = X_2 = X_3 =$ any arbitrary constant k. Thus, one of the eigenvectors of the impedance matrix is

$$\begin{bmatrix} k \\ k \\ k \end{bmatrix}$$

It can be called the zero sequence eigenvector of the symmetrical component transformation matrix and can be written as

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Similarly for $\lambda = Z - M$:

$$\begin{bmatrix} Z-(Z-M) & M & M \\ M & Z-(Z-M) & M \\ M & M & Z-(Z-M) \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = 0$$

which gives

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = 0$$

This gives the general relation $X_1 + X_2 + X_3 = 0$. Any choice of X_1, X_2, X_3 which satisfies this relation is a solution vector. Some choices are shown below:

$$\begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 1 \\ a^2 \\ a \end{bmatrix}, \begin{bmatrix} 1 \\ a \\ a^2 \end{bmatrix}, \begin{bmatrix} 0 \\ \sqrt{3}/2 \\ -\sqrt{3}/2 \end{bmatrix}, \begin{bmatrix} 1 \\ -1/2 \\ -1/2 \end{bmatrix}$$

where a is a unit vector operator, which rotates by 120° in the counterclockwise direction, as defined before.

The symmetrical component transformation is given by solution vectors:

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ a \\ a^2 \end{bmatrix} \begin{bmatrix} 1 \\ a^2 \\ a \end{bmatrix}$$

A symmetrical component transformation matrix can, therefore, be written as

$$\bar{T}_s = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \qquad \bar{T}_s^{-1} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}$$

$$\bar{I}_{abc} = \bar{T}_s \bar{I}_{012} \qquad \bar{I}_{012} = \bar{T}_s^{-1} \bar{I}_{abc}$$

$$\bar{V}_{abc} = \bar{T}_s \bar{V}_{012} \qquad \bar{V}_{012} = \bar{T}_s^{-1} \bar{V}_{abc}$$

$$\bar{V}_{abc} = \bar{Z}_{abc}\bar{I}_{abc}$$

$$\bar{T}_{\mathrm{s}}\bar{V}_{012} = \bar{Z}_{abc}\bar{T}_{\mathrm{s}}\bar{I}_{012}$$

$$\bar{V}_{012} = \bar{T}_{\mathrm{s}}^{-1}\bar{Z}_{abc}\bar{T}_{\mathrm{s}}\bar{I}_{012} = \bar{Z}_{012}\bar{I}_{012}$$

Therefore,

$$\bar{Z}_{012} = \bar{T}_{\mathrm{s}}^{-1}\bar{Z}_{abc}\bar{T}_{\mathrm{s}}$$

$$\bar{Z}_{abc} = \bar{T}_{\mathrm{s}}\bar{Z}_{012}\bar{T}_{\mathrm{s}}^{-1}$$

CLARKE COMPONENT TRANSFORMATION

It has been already shown that, for perfectly symmetrical systems, the component eigenvectors are the same, but eigenvalues can be different. The Clarke component transformation is defined as

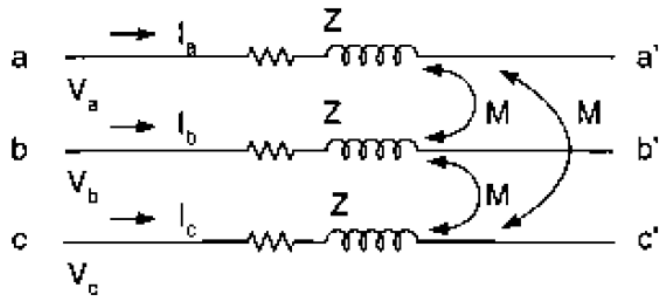
$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ 1 & -\frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} V_0 \\ V_\alpha \\ V_\beta \end{bmatrix}$$

$$\begin{bmatrix} V_0 \\ V_\alpha \\ V_\beta \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ 0 & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}$$

The transformation matrices are

$$\bar{T}_c = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1/2 & \sqrt{3}/2 \\ 1 & 1/2 & \sqrt{3}/2 \end{bmatrix} \qquad \bar{T}_c^{-1} = \begin{bmatrix} 1/3 & 1/3 & 1/3 \\ 2/3 & -1/3 & -1/3 \\ 0 & 1/\sqrt{3} & -1/\sqrt{3} \end{bmatrix}$$

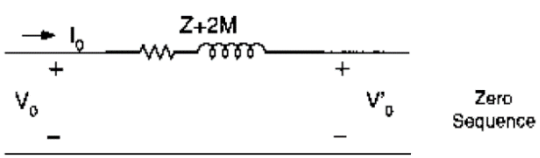
Decoupling a Three-Phase Symmetrical System

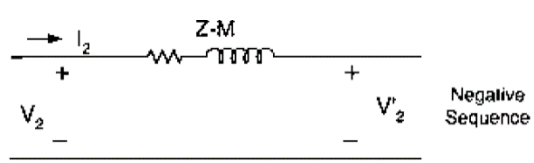


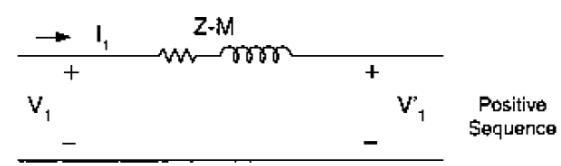
(a)

$$\begin{bmatrix} Z_{aa} & Z_{ab} & Z_{ac} \\ Z_{ba} & Z_{bb} & Z_{bc} \\ Z_{ca} & Z_{cb} & Z_{cc} \end{bmatrix}$$

$$\bar{Z}_{012} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} Z & M & M \\ M & Z & M \\ M & M & Z \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} = \begin{bmatrix} Z+2M & 0 & 0 \\ 0 & Z-M & 0 \\ 0 & 0 & Z-M \end{bmatrix}$$







Decoupling a Three-Phase Unsymmetrical System

$$\bar{Z}_{abc} = \begin{bmatrix} Z_1 & 0 & 0 \\ 0 & Z_2 & 0 \\ 0 & 0 & Z_3 \end{bmatrix}$$

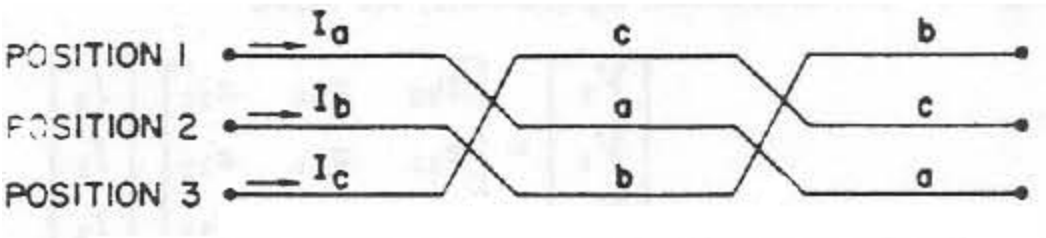
$$\bar{Z}_{012} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} Z_1 & 0 & 0 \\ 0 & Z_2 & 0 \\ 0 & 0 & Z_3 \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix}$$

The resulting matrix shows that the *original unbalanced system is not decoupled.*

$$= \frac{1}{3}\begin{bmatrix} Z_1 + Z_2 + Z_3 & Z_1 + a^2 Z_2 + aZ_3 & Z_1 + aZ_2 + aZ_3 \\ Z_1 + aZ_2 + aZ_3 & Z_1 + Z_2 + Z_3 & Z_1 + a^2 Z_2 + aZ_3 \\ Z_1 + a^2 Z_2 + aZ_3 & Z_1 + aZ_2 + aZ_3 & Z_1 + Z_2 + Z_3 \end{bmatrix}$$

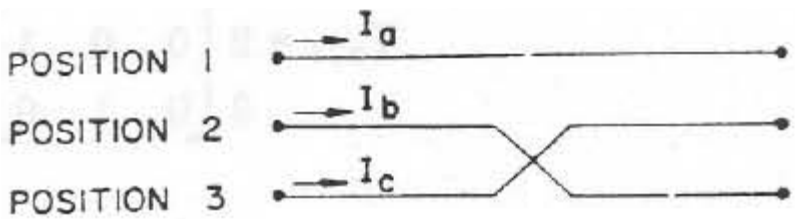
Transposition

Case 1. Rotation

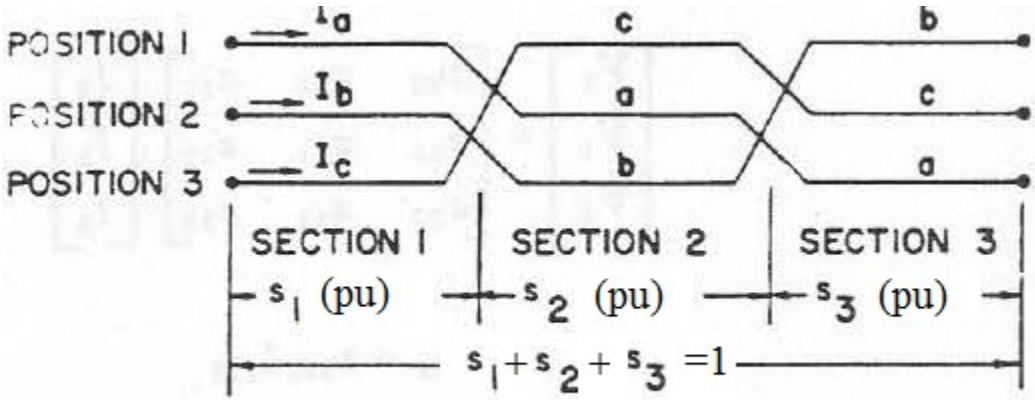


A complete transposition or rotation cycle.

Case 2. Twist



A line with wires 2 and 3 twisted.



A complete transposition or rotation cycle.

$$Z_{abc} = s_1 \times \begin{bmatrix} Z_{11} & Z_{12} & Z_{13} \\ Z_{21} & Z_{22} & Z_{23} \\ Z_{31} & Z_{32} & Z_{33} \end{bmatrix} + s_2 \times \begin{bmatrix} Z_{22} & Z_{23} & Z_{21} \\ Z_{32} & Z_{33} & Z_{31} \\ Z_{12} & Z_{13} & Z_{11} \end{bmatrix} + s_3 \times \begin{bmatrix} Z_{33} & Z_{31} & Z_{32} \\ Z_{13} & Z_{11} & Z_{12} \\ Z_{23} & Z_{21} & Z_{22} \end{bmatrix}$$

Usually $Z_{11} = Z_{22} = Z_{33} = Z_s$ (Same type of a conductor in all phases)

$$Z_{abc} = \begin{bmatrix} Z_s & Z_{k1} & Z_{k2} \\ Z_{k1} & Z_s & Z_{k3} \\ Z_{k2} & Z_{k3} & Z_s \end{bmatrix}$$

$$Z_{k1} = s_1 Z_{12} + s_2 Z_{23} + s_3 Z_{13}$$

$$Z_{k2} = s_1 Z_{13} + s_2 Z_{12} + s_3 Z_{23}$$

$$Z_{k3} = s_1 Z_{23} + s_2 Z_{13} + s_3 Z_{12}$$

$$Z_s = Z_{11} = Z_{22} = Z_{33}$$

Special case S1=S2=S3=1/3

$$Z_{k1} = \frac{1}{3}(Z_{12} + Z_{23} + Z_{13})$$

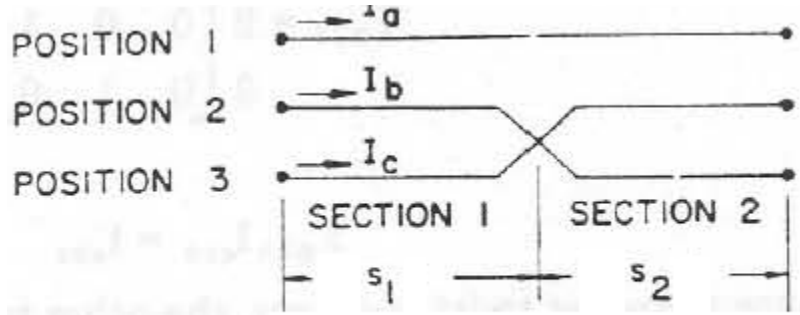
$$Z_{k2} = \frac{1}{3}(Z_{13} + Z_{12} + Z_{23})$$

$$Z_{k3} = \frac{1}{3}(Z_{23} + Z_{13} + Z_{12})$$

Observe that $Z_{k1} = Z_{k2} = Z_{k3} \triangleq Z_m$

$$Z_{abc} = \begin{bmatrix} Z_s & Z_m & Z_m \\ Z_m & Z_s & Z_m \\ Z_m & Z_m & Z_s \end{bmatrix}$$

This case is known as completely transposed line



A line with wires 2 and 3 twisted

$$Z_{abc} = s_1 \times \begin{bmatrix} Z_{11} & Z_{12} & Z_{13} \\ Z_{21} & Z_{22} & Z_{23} \\ Z_{31} & Z_{32} & Z_{33} \end{bmatrix} + s_2 \times \begin{bmatrix} Z_{11} & Z_{13} & Z_{12} \\ Z_{13} & Z_{33} & Z_{23} \\ Z_{12} & Z_{23} & Z_{22} \end{bmatrix}$$

$$Z_{abc} = \begin{bmatrix} Z_s & Z_{k1} & Z_{k2} \\ Z_{k1} & Z_s & Z_{k3} \\ Z_{k2} & Z_{k3} & Z_s \end{bmatrix}$$

Weighted averages

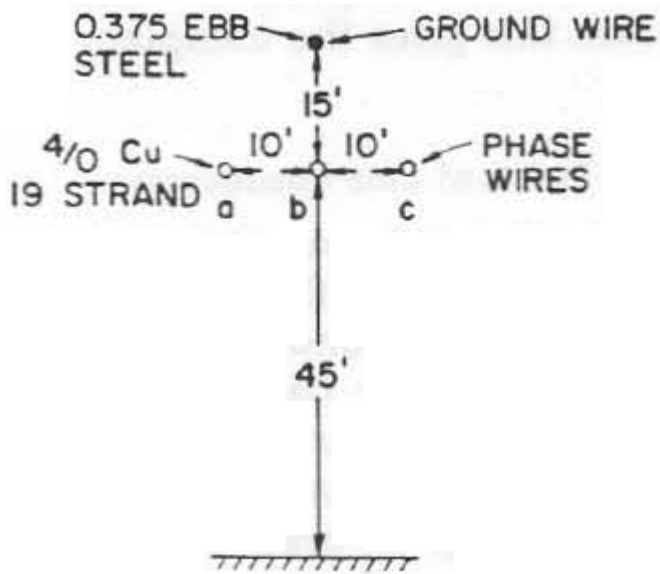
$$Z_{k1} = s_1 Z_{12} + s_2 Z_{13}$$

$$Z_{k2} = s_1 Z_{13} + s_2 Z_{12}$$

$$Z_{k3} = s_1 Z_{23} + s_2 Z_{23} = Z_{23}$$

Observe that $Z_{k1} = Z_{k2} = Z_{k3} \triangleq Z_m$

Example:
For the following configuration



Line configuration of a 69 kV circuit

$$Z_{abc} = \begin{bmatrix} (14.932 + j58.376) & (3.812 + j27.332) & (3.812 + j23.968) \\ (3.812 + j27.332) & (14.932 + j58.376) & (3.812 + j27.332) \\ (3.812 + j23.968) & (3.812 + j27.332) & (14.932 + j58.376) \end{bmatrix} \Omega$$

Conduct:

1) Transposed at S1=0.4, S2=0.4, and S3=0.2
2) Transposed completely S1=S2=S3=1/3
3) Twisted b to c at S1=S2=0.5

1) Transposed at S1=0.4, S2=0.4, and S3=0.2

Zs=14.93+j58.38
Zk1=3.81+j26.66
Zk2=3.81+j25.99
Zk3=3.81+j25.99

Let us compute the differences

Un transposed:
Diff max=|27.33-23.97|=3.36
Transposed:
Diff max=|26.66-25.99|=0.67 (much less!)

2) Transposed completely S1=S2=S3=1/3

Zs=14.93+j58.38
Zk1=Zk2=Zk3=Zm=3.81+j26.21

3) Twisted b to c at S1=S2=0.5

Zs=14.93+j58.38
Zk1=3.81+j25.65
Zk2=3.81+j25.65
Zk3=3.81+j27.33

Diff max=|27.33-25.65|=1.68